Graphic Era
HILL UNIVERSITY
Established by an Act of the State Legislature of Uttarakhand (Adhiniyam Sankhya 12 of 2011)
University under section 2(f) of UGC Act, 1956

End Term (Even) Semester Regular/Back Examination May 2026

Roll no. 24922001....................

Name of the Program and semester: BCA/BCA (AI/DS) VI
Name of the Course: Introduction to Mobile Application Development
Course Code: TBC 602/TBD 603(b)

Time: 3 hours

Maximum Marks: 100

Note:

(i) All the questions are compulsory.
(ii) Answer any two sub questions from a, b and c in each main question.
(iii) Total marks for each question is 20 (twenty).
(iv) Each sub-question carries 10 marks.

Q1.
(2X10=20 Marks)

a. Critically analyze the trade-offs between native, hybrid, and cross-platform mobile development approaches for a large-scale enterprise application. CO1
b. Evaluate the impact of Android's open-source nature on security, customization, and fragmentation. CO2
c. Design a mobile UI layout for a login screen that follows accessibility guidelines (contrast, font size, screen reader support). CO4

Q2.
(2X10=20 Marks)

a. Analyze how different Android versions affect backward compatibility in app development. CO2
b. Propose improvements to Android architecture to enhance performance and security. CO2
c. Differentiate between Activity, Service, Broadcast Receiver, and Content Provider. CO2

Q3.
(2X10=20 Marks)

a. Compare different types of intents and justify their usage in real-world scenarios. CO3
b. Develop an Android application that dynamically switches between activities based on user input. CO3
c. Describe the functionality of basic views like TextView, EditText, and Button. CO4

Q4.
(2X10=20 Marks)

a. Compare ScrollView with RecyclerView and justify when each should be used. CO4
b. Design a responsive UI layout that adapts to different screen sizes and orientations. CO4
c. Describe the role of MainActivity.java and activity_main.xml. CO5

Q5.
(2X10=20 Marks)

a. Critically analyze the directory structure of Android applications in terms of modularity and maintainability. CO5
b. Compare different storage mechanisms (Shared Preferences, Files, SQLite) based on performance and use cases. CO6
c. Evaluate the role of Toast messages in user feedback and suggest better alternatives. CO6